Writing Equations

Exam code: 0620 & 0971

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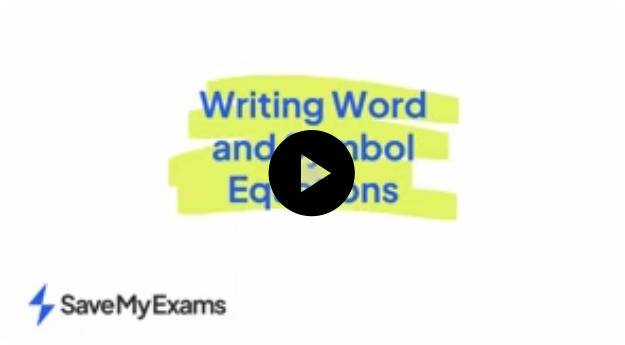
Updated on 29 September 2024



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Writing word equations & symbol equations

Word equations

- Word equations show the **reactants** and **products** of a chemical reaction using their full chemical names

reactants → products

- The reactants are the substances on the **left-hand side** of the arrow
  - They can be thought of as the chemical **ingredients** of the reaction
- They react with each other to form new substances, which are the products
- The products are on the **right-hand side** of the arrow
- The arrow (which is spoken as "*to form*" or "*produces*") implies the **conversion** of reactants into products
  - Reaction **conditions** or the name of a **catalyst** (a substance added to make a reaction go faster) can be written above the arrow
- An example is the reaction of sodium hydroxide (a base) and hydrochloric acid to produce sodium chloride (common table salt) and water:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

1. Ammonia reacts with nitric acid to form the fertiliser ammonium nitrate. Write a word equation for the reaction taking place.
2. Iron(II) hydroxide and sodium sulfate are formed when iron(II) sulfate solution and sodium hydroxide react together. Write a word equation for the reaction taking place.
3. Carbon is the main element found in coal and burns in air to produce carbon dioxide. Write a word equation for the reaction taking place.

Answers:

1. Ammonia + nitric acid → ammonium nitrate
   - This question has all the information in the correct order
   - *Ammonia reacts with nitric acid*
     - This becomes ammonia + nitric acid
   - *to form*
     - This is the arrow in the equation
   - *to form the fertiliser ammonium nitrate*
     - This tells you that the product is ammonium nitrate
2. Iron(II) sulfate + sodium hydroxide → iron(II) hydroxide + sodium sulfate
   - **Careful:** This question has all the required information but the products are written first
   - *Iron(II) hydroxide and sodium sulfate are formed*
     - This becomes → iron(II) hydroxide + sodium sulfate
   - *when iron(II) sulfate solution and sodium hydroxide react together*
     - This becomes Iron(II) sulfate + sodium hydroxide →
3. Carbon + oxygen → carbon dioxide
   - **Careful:** Not all of the required information is given in the question
   - You are expected to know that burning in air means that the chemical is reacting with oxygen
   - *Carbon... ...burns in air*
     - This becomes carbon + oxygen
   - *to produce*
     - This is the arrow in the equation
   - *to produce carbon dioxide*
     - This tells you that the product is carbon dioxide

Help me understand

Symbol equations

- A **symbol equation** uses the formulae of the reactants and products to show what happens in a chemical reaction
- When writing symbol equations, you should:
  - Ensure reactants are on the left of the equation and products are on the right

        - Solid = (s)
        - Liquid = (l)
        - Gas = (g)
        - Aqueous = (aq)

- Sometimes it can be hard to know what the correct state symbol is and we have to look for clues in the identity of substances in a reaction
- Generally, unless they are in a solution:
    - Metal compounds will always be solid, although there are a few exceptions
    - Ionic compounds will usually be solids

- Non-metal compounds can be solids, liquids or gases
    - So, it depends on information given

- Precipitates formed in solution count as solids
- A symbol equation must be balanced to give the correct ratio of reactants and products:
    - For example, the combustion of sulfur:

$$S\,(s) + O_2\,(g) \rightarrow SO_2\,(g)$$

- This equation shows that one atom of solid sulfur, S, reacts with one gaseous molecule of oxygen, O_2, to make one gaseous molecule of sulfur dioxide, SO_2

Examiner Tips and Tricks

- In exams, you will not need to include them in all equations unless you are specifically asked to .
    - However, it is good practice to include state symbols in your equations so that you don't miss any marks.
- Be careful when writing the state symbol of solutions of liquids.
    - For example, ethanol, or common alcohol, is a liquid at room temperature, so if it is pure alcohol then you would be using (l) as the state symbol.
    - However, most of the time alcohol is used as a solution in water so the state symbol should be (aq).

Balancing symbol equations

- When balancing equations, there must be the **same number of atoms** of each element on either side of the equation following the law of conservation of mass
- To balance an equation you work across the equation from left to right, checking one element after another
    - If there is a group of atoms such as a nitrate group (NO_3^-) that has not changed from one side to the other, then count the whole group as one entity rather than counting the individual atoms

- Examples of balanced symbol / chemical equations include:
    - Acid-base neutralisation reaction:

$$2Fe_2O_3\,(aq) + 3C\,(s) \longrightarrow 4Fe\,(s) + 3CO_2\,(g)$$

- In each equation, there are equal numbers of each atom on either side of the reaction arrow so the equations are balanced
- The best approach is to practice lot of examples of balancing equations
- This can be by trial and error - changing the coefficients (numbers) in front of the formulae one by one and checking the result on the other side
- Balance elements that appear on their own, last in the process

Worked Example

When magnesium oxide, MgO, reacts with nitric acid, HNO_3, it forms magnesium nitrate, $Mg(NO_3)_2$, and water.

magnesium oxide + nitric acid ⟶ magnesium nitrate + water

Write the balanced **symbol** equation for this reaction.

Answer:

- The balanced symbol equation is:

$$MgO\,(s) + \mathbf{2}HNO_3\,(aq) \longrightarrow Mg(NO_3)_2\,(aq) + H_2O\,(l)$$

- **Step 1** - writing the unbalanced equation
  - Magnesium oxide, MgO, reacts with nitric acid, HNO_3, it forms magnesium nitrate, $Mg(NO_3)_2$, and water
    - $MgO + HNO_3 \longrightarrow Mg(NO_3)_2 + H_2O$
  - The Mg and O atoms (not including the O in the NO_3 group appear to be balanced), so we should focus on the H atoms and NO_3 groups
- **Step 2** - balancing hydrogen atoms
  - There are 2 hydrogen atoms on the product side, so 2 hydrogen atoms are needed on the reactant side
  - This means that $2HNO_3$ will be needed as we cannot change the chemical formula
    - $MgO + \mathbf{2}HNO_3 \longrightarrow Mg(NO_3)_2 + H_2O$
  - This also balances the nitrate, NO_3, groups
- **Step 3** - checking the equation
  - The equation appears balanced so we need to check that it is:
  - Reactant side:
    - Mg atom
    - 1 O atom - not including those in the NO_3 group
    - 2 H atoms
    - 2 NO_3 groups - remember to keep groups as a single entity if they are unchanged on both sides of the equation
  - Product side:
    - 1 Mg atom
    - 2 NO_3 groups - remember to keep groups as a single entity if they are unchanged on both sides of the equation
    - 2 H atoms

Worked Example

Aluminium reacts with copper(II) oxide to produce aluminium oxide and copper. Balance the symbol equation for the reaction taking place.

$$_Al\,(s) + _CuO\,(s) \longrightarrow _Al_2O_3\,(s) + _Cu\,(s)$$

Answer:

- The balanced symbol equation is:

$$\mathbf{2}Al\,(s) + \mathbf{3}CuO\,(s) \longrightarrow Al_2O_3\,(s) + \mathbf{3}Cu\,(s)$$

- Step 1 - balancing aluminium atoms
  - There are 2 aluminium atoms on the product side, so 2 aluminium atoms are needed on the reactant side
    - $\mathbf{2}Al + CuO \longrightarrow Al_2O_3 + _Cu$
- Step 2 - balancing oxygen atoms
  - There are 3 oxygen atoms on the product side, so 3 oxygen atoms are needed on the reactant side
  - This means that 3 CuO will be needed as we cannot change the chemical formula
    - $\mathbf{2}Al + \mathbf{3}CuO \longrightarrow Al_2O_3 + Cu$
- Step 3 - balancing copper atoms
  - There are 3 copper atoms on the reactant side, so 3 copper atoms are needed on the product side
    - $\mathbf{2}Al + \mathbf{3}CuO \longrightarrow _Al_2O_3 + \mathbf{3}Cu$
- The equation is now balanced

Deducing symbol equations

Extended tier only

- For some reactions, you will not be given the unbalanced equation
  - You will be expected to know or deduce the formula of compounds and then balance the equations

Worked Example

Aluminium burns in chlorine to form the white solid, aluminium chloride.

Write the balanced symbol equation, including state symbols, for the reaction.

Answer:

an element so its formula is the same as its chemical symbol: **Al (s)**
    - From your knowledge of Group VII elements, you should know that chlorine is a gas that exists as a diatomic molecule: **Cl_2 (g)**
    - Aluminum chloride is a solid - this information is given in the question as you would not be expected to know this.
        - Its formula is deduced from the charges on the ions present:
        - Aluminium has a 3+ charge
        - Chloride ions have a 1– charge
        - Therefore, for the compound to be neutral, 3 chloride ions are needed for every 1 aluminium ion: **$AlCl_3$ (s)**
2. Construct an unbalanced symbol equation:
    - The unbalanced symbol equation is:
        - $Al\,(s) + Cl_2\,(g) \rightarrow AlCl_3\,(s)$
3. Balance the equation:
    - Make the number of Cl atoms on the right-hand side an even number by adding a 2 in front of $AlCl_3$:
        - $Al\,(s) + Cl_2\,(g) \rightarrow \mathbf{2}AlCl_3\,(s)$
    - This gives 6 Cl atoms on the right-hand side
    - So, now balance the number of Cl atoms, on the left-hand side, by adding a 3 in front of Cl_2:
        - $Al\,(s) + \mathbf{3}Cl_2\,(g) \rightarrow \mathbf{2}AlCl_3\,(s)$
    - This gives 2 Al atoms on the right-hand side
    - So, balance the number of Al atoms, on the left-hand side, by adding a 2 in front of the Al:
        - $\mathbf{2}Al\,(s) + \mathbf{3}Cl_2\,(g) \rightarrow \mathbf{2}AlCl_3\,(s)$

Examiner Tips and Tricks

When balancing equations you **cannot** change any of the formulae, only the amount of each atom or molecule. This is done by changing the numbers that go in **front** of each chemical species.

Balancing Ionic Equations

- In aqueous solutions, ionic compounds **dissociate** into their ions
    - This means that they separate into their component ions
- For example, hydrochloric acid and potassium hydroxide dissociate as follows:

$$\mathbf{HCl\,(aq) \rightarrow H^+\,(aq) + Cl^-(aq)}$$

$$\mathbf{KOH\,(aq) \rightarrow K^+\,(aq) + OH^-\,(aq)}$$

- It is important that you can recognise common ionic compounds and their constituent ions
    - These include:

- Soluble salts e.g. potassium sulfate, sodium chloride

- The steps to writing an **ionic equation** are:
  1. Write the full, balanced symbol equation
  2. Replace the ionic compounds in the balanced symbol equation with the component ions
  3. Remove any ions that appear on both sides of the equation

Worked Example

Write the ionic equation for the displacement reaction of aqueous chlorine and aqueous potassium iodide.

Answer:

1. Write out the full balanced equation:
   - $2KI(aq) + Cl_2(aq) \rightarrow 2KCl(aq) + I_2(aq)$
2. Replace the ionic compounds in the balanced symbol equation with the component ions
   - $2K^+(aq) + 2I^-(aq) + Cl_2(aq) \rightarrow 2K^+(aq) + 2Cl^-(aq) + I_2(aq)$
3. Remove any ions that appear on both sides of the equation:
   - $2I^-(aq) + Cl_2(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$

Examiner Tips and Tricks

Ionic equations should **always** have state symbols included.

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